

Assignment 2 Prompt 3 Itzel Polin Lopez

CUSTOMER_ADDRESS	
PK customer_id	-- also FK → CUSTOMER.customer_id
street_address	
city	
province_state	
postal_code	
country	

Architecture that overwrites the address

- There is **one row per customer** in CUSTOMER_ADDRESS.
- When a customer moves, the existing row is updated with the new address.
- The old address is **not retained** anywhere.

Slowly Changing Dimension Type 1:

Changes are applied in place, and history is not kept.

CUSTOMER_ADDRESS	
PK customer_address_id	-- surrogate key
FK customer_id	-- → CUSTOMER.customer_id
street_address	
city	
province_state	
postal_code	
country	
start_date	-- when this address became valid
end_date	-- when it stopped being valid (NULL if current)
is_current_flag	-- indicates the active address (Y/N or 1/0)

Architecture that retains the address

- A customer can have **multiple rows** over time in CUSTOMER_ADDRESS.
- When the address changes:
- The previous row's end_date is set and is_current_flag is turned off.
- A new row is inserted with the new address, start_date, end_date = NULL, and is_current_flag = 1.
- This preserves a complete address history for auditing or analysis.

Slowly Changing Dimension Type 2:

Changes are modeled by inserting new rows with effective dates so history is retained

If the address is only needed for current deliveries and basic contact information, then a Type 1 overwrite is simpler and avoids unnecessary data. However, if the store wants to analyze customer movement over time (for example, to understand shipping trends, relocating customers, or long-term loyalty), then a Type 2 design is better because it keeps a full historical record of address changes